

Chapter 99 — Prevention of Urinary Stone Disease

INTRODUCTION

Stone disease as a chronic condition

- From the prevention perspective, urinary stone disease (USD) is best conceptualised as a **chronic metabolic condition punctuated by periodic stone attacks**, rather than as a series of isolated colic episodes.
- It is characterised by **high recurrence**, association with **predisposing chronic conditions** (diabetes, obesity, gout) and **identifiable metabolic disturbances** such as hypocitraturia.
- Preventive practice follows the **American Urological Association (AUA)** and **European Association of Urology (EAU)** guidelines.

Recurrence and its prediction

- Recurrence risk is **up to 50% within 5 years**, but varies greatly between patients.
- The **Recurrence of Kidney Stone (ROKS) nomogram** predicts the risk of a second stone from demographic and clinical factors; 10-year risk ranges from about 12% in the lowest to 56% in the highest risk stratum.
- **ROKS 2** predicts **multiple recurrent symptomatic episodes** (highest-risk group: four or more). Risk factors: **younger age, male sex, higher body mass index (BMI), family history, pregnancy**, an asymptomatic or suspected stone before the first episode, a history of **brushite, struvite or uric acid** stone, and radiographic factors.
- Where urine chemistry was available, **low urine volume and low citrate** predicted recurrence; predicted 5-year recurrence spans almost the full range (about 1% to 94%), allowing personalised counselling.

Link with the metabolic syndrome

- National Health and Nutrition Examination Survey (NHANES) data show strong associations between **obesity, diabetes, gout** and lifetime stone risk.
- **Gout lowers urine pH**, favouring uric acid stones; **diabetes raises urinary calcium** and also lowers urine pH.

Prevalence, incidence and sex distribution

- Prevalence is rising rapidly: about **1 in 11** people in the United States form a stone in their lifetime, nearly double the figure of 15 years earlier; recent 12-month incidence exceeds historical estimates.
- Historically male-predominant, the disease is now **close to equal between the sexes** (male:female about 1.2–1.45:1), because of a disproportionate **rise in female (especially younger female) stone formation**, not a fall in male stone formation.

MORBIDITY

- The burden comprises **severe colic, emergency care, lost work and life activity**, and for many, surgery; about one-third of employees treated for stones miss work.
- First-time stone formers carry about a **50% risk of recurrence within the subsequent 10 years**.

QUALITY-OF-LIFE IMPACT

- Recurrent stone formers are especially affected, through **chronic or episodic pain, repeated clinic visits and interventions**, and side effects of preventive drugs or diets.
- On the **Short Form Health Survey (SF-36)**, health-related quality of life (QoL) is below the population mean, and **worse in females** than males.
- The **Patient-Reported Outcomes Measurement Information System (PROMIS)**, supported by the National Institutes of Health, shows worse pain intensity, pain interference and physical function than the general population across phases of a stone episode.
- A validated **disease-specific** QoL instrument has shown worse QoL in **cystine** than non-cystine formers, and in patients with **asymptomatic stones** than in those with no stones.

KEY POINTS

- The prevalence and incidence of USD are increasing.
- Prevalence is increasing rapidly in females and non-White populations.
- USD is a chronic metabolic condition, with recurrent events in up to 50% of affected individuals within 5 to 10 years.
- USD negatively affects functioning and QoL.

Concept — *stone disease is a rising, recurrent, systemic metabolic disorder allied to the metabolic syndrome; the colic is the symptom, recurrence is the disease, and prevention is therefore the core long-term task.*

SCREENING EVALUATION FOR NEW STONE FORMERS (BOX 99.1)

- AUA and EAU recommend a **screening evaluation for every first-time stone former**, to identify those at **high risk of recurrence** from routine history, blood tests, urinalysis and stone analysis.
- **History**: comorbidities, medications and family history. The EAU asks for a **holistic** assessment — the likelihood of recurrence and also its **consequences** for that individual (e.g., chronic kidney disease, mineral bone disorder). Eliciting **gastrointestinal (GI) disease and sarcoidosis** is critical. Diet is reviewed: calcium, sodium, fluid, protein, fruit and vegetables, oxalate-rich foods and supplements such as vitamin C.
- **Children** who form stones are generally high risk and should have metabolic evaluation.
- **Blood tests** look for an underlying cause — for example, **hypercalcaemia suggesting hyperparathyroidism**.
- **Stone analysis** points to the cause: **uric acid** suggests a metabolic condition, **cystine** a genetic one, and **struvite** in recurrent urinary tract infection (UTI) confirms infection as the cause and recurrence risk.
- **Imaging** quantifies **stone burden**, which predicts future symptomatic events.

BOX 99.1 SCREENING EVALUATION OF NEW STONE FORMERS

Component	What is assessed
Medical history	GI disease, gout, recurrent UTI, mineral bone density disease, family history; medications — calcium, vitamin C, vitamin D, acetazolamide, steroids
Blood tests	Sodium, potassium, creatinine, calcium, uric acid
Urine	pH, crystalluria; culture if infection suggested
Stone analysis	Types pointing to a predisposing condition — cystine, struvite, uric acid

Component	What is assessed
Radiography	Baseline stone burden

- **Purpose of the box:** a short, inexpensive triage that sorts first-time formers into high and low risk; **24-hour urine is not part of screening.**
- **Trap:** the medication history includes **calcium, vitamins C and D, acetazolamide and steroids** — each a lithogenic drug.

KEY POINTS

- First-time stone formers should undergo a screening evaluation to be classified as high or low risk for stone recurrence.
- Medical history, blood and urine testing, and stone composition are recommended by current AUA and EAU guidelines.

Concept — *the screening evaluation is a triage step: its job is to sort first-time formers into high risk (proceed to 24-hour urine) and low risk (general advice only).*

GENERAL RECOMMENDATIONS FOR STONE PREVENTION IN NEW STONE FORMERS

- Dietary and lifestyle advice in both guidelines — particularly important for **low-risk** patients who will not undergo metabolic evaluation — covers fluid, protein, fruit and vegetables, sodium, calcium and oxalate.

Fluid

- **Low urine volume is perhaps the most important single risk factor;** higher fluid intake prevents recurrence of **all stone types**, probably by lowering urinary supersaturation.
- **Borghi et al. (1996)**, the landmark randomised controlled trial (RCT) in idiopathic calcium oxalate formers: fluid sufficient to keep **urine volume above 2 L/day** reduced 5-year recurrence (12.1% vs 27%).
- The **Health Professionals Follow-Up Study (HPFS)** and **Nurses' Health Study (NHS) I** cohorts confirm lower stone incidence with fluid intake above 2.5 L/day.
- **Guideline target: fluid sufficient for urine output of at least 2.5 L/day;** in practice about **100 oz (about 3 L) daily**. There is no strong evidence for targets tailored to body habitus or sex.

Adherence

- **Adherence is poor:** counselled patients raise urine volume only modestly, and only about half reach 2.5 L/day at 6 months.
- **Mobile health technology** (applications, fitness trackers, social media) supports behaviour change through **goal setting, self-monitoring and feedback;** short-term **smart water bottle** studies suggest increased intake, with longer trials in progress.

Animal protein

- Nondairy animal protein (meat, fish, poultry, eggs) **raises urinary calcium** through bone resorption and reduced tubular calcium reabsorption.
- Its **purine and acid load** reduces tubular citrate reabsorption, causing **hypocitraturia**, and raises urinary uric acid, causing **hyperuricosuria**. Its effect on urinary oxalate is conflicting.

- The harm is greatest when a high animal-protein diet is **combined with low calcium intake**. Short-term protein restriction in hypercalciuric formers improves urinary calcium, uric acid, oxalate and citrate.
- Clinical-trial support is thin: both guidelines endorse limiting animal protein — especially in **uric acid** formers and **calcium oxalate formers with high urinary uric acid** — on **expert opinion**. A 4-year RCT of low-animal-protein or high-fibre diets showed no reduction in urinary calcium or recurrence.
- **Borghi et al. (2002)**, landmark RCT in hypercalciuric formers: a **low-protein, low-salt, normal-calcium diet** halved stone events compared with a **low-calcium diet**, although the three components cannot be separated.

Sodium, calcium, fruit and vegetables

- The **Dietary Approaches to Stop Hypertension (DASH)** diet is associated with lower stone risk; **low sodium** is recommended because sodium raises calcium excretion and lowers urinary citrate.
- **Calcium restriction increases stone risk**; a diet high in **fruit and vegetables** improves urine parameters.
- These recommendations are guideline-endorsed but need stronger randomised evidence.

KEY POINTS

- Fluid intake to maintain urine output of 2.5 L/day is the cornerstone of dietary prevention of urinary stone recurrence.
- A diet low in animal protein and high in fruits and vegetables is recommended by current guidelines, although randomised trial evidence is limited.
- Guidelines also support low sodium and moderate calcium intake, along with consideration of a low-oxalate diet.

Concept — *dilution is universal and evidence-based; the dietary levers (less animal protein and salt, normal calcium, more fruit and vegetables) act by lowering calcium and uric acid and raising citrate — mostly on physiological and expert-opinion grounds.*

METABOLIC EVALUATION FOR HIGH-RISK AND RECURRENT STONE FORMERS

- On expert opinion, metabolic testing is **one or two 24-hour urine collections on a random diet**, analysed at minimum for **volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium and creatinine**.
- One versus two collections is **unresolved**: one retrospective comparison found two random-diet collections similar enough that one suffices; a larger series found clinically relevant differences between paired collections in most comparisons, favouring two.
- **Fast and calcium-load testing** to subclassify hypercalciuria is **discouraged**, because it does not change management.
- **Real-world uptake is low** — few eligible patients have a 24-hour urine, and few on drug therapy complete the recommended follow-up tests; yet claims data show that **targeted drug therapy matched to 24-hour urine findings** significantly reduces symptomatic stone events.

KEY POINTS

- High-risk stone formers include those with GI disease, obesity, metabolic syndrome, diabetes mellitus, primary hyperparathyroidism, renal tubular acidosis, sarcoidosis, gout, and family history of nephrolithiasis.

- Children should generally be evaluated because of concerns about renal damage and long-term sequelae of stone recurrence.
- High-risk individuals should be offered metabolic evaluation, including at least one properly collected 24-hour urine collection, to assess for stone-related risk factors.

Concept — *the 24-hour urine converts non-specific advice into targeted therapy, and targeted therapy works — yet it is badly underused in practice.*

PHARMACOLOGIC PREVENTION STRATEGIES

Hypercalciuria

Thiazides and Thiazide-Like Diuretics

- Thiazides are the **preferred drug for hypercalciuria** and among the best studied in stone prevention, effective in both specific and unselected groups of stone formers.
- The AUA recommends them for **hypercalciuria with recurrent calcium stones**, and for **recurrent calcium stones despite a normal metabolic evaluation**.
- First used by **Yendt (1966)** for undifferentiated hypercalciuria. **Indapamide** is a thiazide-like diuretic, also effective.
- **Mechanism:** direct stimulation of **distal nephron calcium reabsorption** while promoting sodium excretion; with long-term use, **volume contraction** adds proximal tubular reabsorption of sodium and calcium.
- **Ettinger et al. (1988)**, double-blind RCT: chlorthalidone reduced predicted stone formation by about 90% versus placebo or magnesium hydroxide. A second double-blind RCT with hydrochlorothiazide and a Cochrane review both confirm protection against recurrence.

Potassium and electrolytes

- Effects on urinary magnesium and zinc are inconsistent.
- Chain of harm: **potassium loss → hypokalaemia with intracellular acidosis → hypocitraturia**. Hence potassium is supplemented as **potassium citrate or potassium chloride**; adding potassium citrate prevents hypokalaemia and hypochloremic metabolic acidosis during long-term therapy.
- Check **serum potassium 1–2 weeks after starting or changing the dose**; take particular care with known potassium deficiency, digitalis therapy or developing hypocitraturia.

Potassium-sparing combinations

- **Amiloride plus thiazide (Moduretic)** may lower calcium excretion more than thiazide alone but does **not raise citrate**; because amiloride spares potassium, **potassium supplements should not be given**.
- **Triamterene** is used with caution because of **triamterene stones**.

Bone

- Thiazides **protect against osteopenia**: combined with potassium citrate and diet, stone formation fell and bone mineral density rose; a Cochrane review found **fewer hip fractures** on thiazides.

Side effects

- Usually mild at low dose and improve with time, yet **about one in six stop therapy** (Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial [ALLHAT]).
- Thirst, polyuria, cramps and GI upset; thiazides may **precipitate gout or diabetes** by raising serum uric acid and glucose; a small percentage develop reduced libido or sexual dysfunction.

TABLE 99.1 DOSAGES OF COMMON MEDICATIONS USED TO PREVENT URINARY CALCULI AND TABLE 99.2 POTENTIAL SIDE EFFECTS OF MEDICATIONS USED TO PREVENT URINARY LITHIASIS (COMBINED)

Group	Drug	Dose	Side effects
Thiazides	Hydrochlorothiazide	25 mg orally twice daily	Potassium wasting, muscle cramps, hyperuricosuria, intracellular acidosis, hypocitraturia
	Chlorthalidone	25–50 mg daily	
	Indapamide	2.5 mg daily	
Phosphate agents	Sodium cellulose phosphate (SCP)	10–15 g/day divided with meals	SCP: GI distress, hypomagnesaemia, hyperoxaluria, parathyroid hormone stimulation. Orthophosphate: as SCP, plus soft-tissue calcification
	Orthophosphate	0.5 g three times daily	
Alkali	Potassium citrate	20 mEq two to three times daily	GI upset, hyperkalaemia
Uric acid	Allopurinol	300 mg daily	Rash, myalgia
Oxalate	Magnesium gluconate	0.5–1 g three times daily	Magnesium: diarrhoea. Pyridoxine: ataxia, painful skin lesions, heartburn, nausea, photosensitivity, numbness, reduced sensation of pain and extreme temperature
	Pyridoxine (vitamin B6)	100 mg daily	
Cystine (thiols)	D-Penicillamine α -Mercaptopropionylglycine (tiopronin)	250 mg daily, titrated Table: 100 mg twice daily, titrated; text: start 300 mg twice daily, up to 1200 mg/day	D-Penicillamine: nephrotic syndrome, dermatitis, pancytopenia. Tiopronin: rash, asthenia, rheumatological complaints, GI distress, mental status changes
Struvite	Acetohydroxamic acid (AHA)	250 mg two to three times daily (text: three times daily)	Thromboembolism, tremor, headache, palpitations, oedema, GI distress, loss of taste, rash, alopecia, anaemia, abdominal pain

- **What the table is for:** one drug (or class) per metabolic target — calcium, citrate, uric acid, oxalate, cystine, urease — so the 24-hour urine abnormality selects the row.
- **Pattern:** each drug's side effects follow from its chemistry — thiazides waste potassium (hence hypocitraturia), potassium citrate risks hyperkalaemia, thiols carry immunological and haematological toxicity, AHA carries the heaviest burden.
- **Tiopronin dose:** the table gives a starting dose of 100 mg twice daily; the text gives 300 mg twice daily titrated to urinary cystine below 250 mg/L. The source does not reconcile them; both are titrated to effect.
- **Trap:** thiazide side effects include **hyperuricosuria and hypocitraturia** — both lithogenic.

Concept — *thiazides lower urinary calcium by distal calcium reabsorption and volume contraction; their main liability, potassium loss causing hypocitraturia, is anticipated by pairing them with potassium citrate — never with a potassium-sparing agent plus potassium.*

Hyperoxaluria

- One of the **commoner abnormalities** on metabolic evaluation. There is **no threshold**: stone risk rises continuously with urinary oxalate, even within the "normal" range.
- Oxalate comes from diet and metabolism and has **no known function** in humans; susceptibility to dietary oxalate absorption varies, and heritable sensitivity is debated (Chapter 95).

- **Deficiency of *Oxalobacter formigenes*** in the gut flora is increasingly implicated in calcium oxalate stones; it may be **iatrogenic**, as the organism is sensitive to many common **antibiotics**. Recolonisation is under study.

Enteric Hyperoxaluria

- Urinary oxalate **above 40 mg/day** arising from **GI malabsorption and diarrhoea**; often well over 60–70 mg/day, with urinary calcium often well under 100 mg/day.
- Mechanism: malabsorbed fat **saponifies dietary calcium**, leaving oxalate free for absorption; stone risk is added by dehydration, hypokalaemia, bicarbonate loss and hypocitraturia.
- **Roux-en-Y gastric bypass (RYGB)**, a malabsorptive procedure, is an increasingly common cause: urinary oxalate rises and urinary citrate and volume fall, and stone risk exceeds that of matched obese controls. **Restrictive** procedures (band, sleeve) add less direct risk, mainly through dehydration.

Primary Hyperoxaluria

- Primary hyperoxaluria (PH) is an extremely rare **inborn error of glyoxylate metabolism** with three genetic types (see table).
- Presents in **childhood** with early stones, **oxalosis** (tissue deposition) and frequently **renal failure from nephrocalcinosis**; untreated, death may occur before age 20. Urinary oxalate is often **above 100 mg/day**, with high serum oxalate. Subtypes overlap clinically.

PRIMARY HYPEROXALURIA TYPES

Type	Defective enzyme (gene)	Inheritance	Share of PH	Distinguishing feature
PH1	Alanine glyoxylate aminotransferase (AGXT)	Autosomal recessive	About 70%	Commonest; target of pyridoxine and lumasiran
PH2	D-glycerate dehydrogenase (GRHPR)	—	About 10%	Less common
PH3	4-Hydroxy-2-oxoglutarate aldolase (HOGA1)	Autosomal recessive	About 10%	Milder ; lower oxalate, less renal failure and oxalosis; often concomitant hypercalciuria
Unidentified	Unknown	—	About 10%	—

ENTERIC VERSUS PRIMARY HYPEROXALURIA

	Enteric	Primary
Cause	Malabsorption, diarrhoea, RYGB	Inborn error — hepatic overproduction
Urinary oxalate	Above 40 mg/day; often 60–70+	Often above 100 mg/day; high serum oxalate
Urinary calcium	Low	High in PH3
Onset	Acquired, adult	Childhood
Treatment	Calcium with meals, low-oxalate diet, fluid, liquid potassium citrate, magnesium	Pyridoxine; lumasiran (PH1)

DIETARY MANAGEMENT FOR HYPEROXALURIA

- Although oxalate cannot be avoided entirely, **recognising and avoiding high-oxalate foods is a cornerstone** of management.
- Diet contributes a variable share of urinary oxalate — a minority on a low-oxalate diet, rising substantially on a high-oxalate diet — and the **dietary share rises when calcium intake falls**.
- Avoid large portions of **spinach, beets, chocolate, nuts and tea**.
- General advice suits recurrent formers; a strict low-oxalate diet is **most useful in enteric hyperoxaluria, bowel disease and after gastric bypass**.

MEDICAL THERAPY FOR HYPEROXALURIA

- For idiopathic hyperoxaluria, **oxalate avoidance and correction of underlying disease** are the cornerstones; drugs are occasionally useful.

Medical Therapy: Enteric Hyperoxaluria

- **No RCTs** exist.
- **Calcium supplementation is the primary measure**: up to **1 g/day in divided doses, taken with meals** so that it binds oxalate in the gut. **Calcium citrate** also raises urinary citrate and pH through its alkali load. **Very high fluid intake** must accompany it, or calcium may raise overall stone risk.
- **Cholestyramine**, a bile acid sequestrant, should reduce colonic irritability and oxalate hyperabsorption in principle and works in animals, but **human results are disappointing**.
- **Magnesium** binds oxalate and destabilises calcium oxalate crystals; malabsorption causes hypomagnesuria and more free oxalate. Supplements (gluconate, oxide, hydroxide) correct it, but **may worsen diarrhoea**; optimal dosing is unknown, and it may complement potassium citrate.
- **Fluid above 2 L/day at minimum** offsets GI losses; **potassium citrate** corrects hypokalaemia and acidosis — in **liquid form** when transit is rapid, as wax-matrix tablets are poorly absorbed.

Medical Therapy: Primary Hyperoxaluria

- **Pyridoxine (vitamin B6)** acts as a **cofactor**, increasing the expression, activity and peroxisomal import of the PH1 enzyme; with low-oxalate counselling it also lowered urinary oxalate in **idiopathic** hyperoxaluria without significant side effects.
- **Oral O. formigenes** showed early promise, but longer studies with modified formulations **failed to replicate** it.
- **Lumasiran**, approved by the United States Food and Drug Administration (FDA) for PH1, uses **RNA interference against glycolate oxidase messenger RNA**, reducing hepatic oxalate production.
- **ILLUMINATE-A** (phase 3, PH1, 2:1 randomisation, subcutaneous): at 6 months plasma oxalate fell substantially versus placebo, and **84% on lumasiran versus 0% on placebo** reached 24-hour urinary oxalate no more than 1.5 times the upper limit of normal. **ILLUMINATE-B** (children under 6 years): rapid, sustained fall in spot urinary oxalate:creatinine ratio.

Concept — *urinary oxalate risk is continuous; enteric disease is a gut-binding problem (restore luminal calcium–oxalate binding with calcium at meals), whereas primary hyperoxaluria is a hepatic overproduction problem (supply the cofactor or silence the upstream enzyme).*

HYPOCITRATURIA

Pharmacologic Intervention for Hypocitraturia

- Mechanism: oral citrate is **metabolised in the liver to bicarbonate**; the alkali load **inhibits proximal tubular citrate reabsorption**, raising urinary citrate.
- **Potassium citrate** is standard in the United States (wax tablet, liquid, powder), **30–60 mEq/day** divided or as a single evening dose. **Sodium citrate** is used more in Europe or when hyperkalaemia is a concern, but **does not lower urinary calcium**, probably because of its sodium load.
- AUA: offer potassium citrate to **recurrent calcium formers with low citrate**, and — like thiazides — to **recurrent calcium formers without any abnormality**.
- **Evidence**: long-term therapy durably raises urine pH and citrate within about 6 months and lowers stone formation. In RCTs, **Ettinger et al. (1997)** — potassium magnesium citrate, relative risk of a new stone about 0.16 at 3 years — and **Barcelo et al. (1993)** — potassium citrate, markedly fewer new stones at 3 years; a Cochrane review confirms a large reduction.
- **Side effects**: GI upset is commonest; take with water after meals. The wax tablet **shell seen in stool is normal** and does not mean the drug was not absorbed. More trial dropouts occur for adverse events. **Monitor potassium in renal insufficiency and avoid with potassium-sparing diuretics**.
- **Urine pH**: alkalinisation helps calcium oxalate formers (inhibitors work better in alkaline urine), but **pH of 7 or above raises calcium phosphate stone risk**. The calcium phosphate content of stones has risen over three decades with rising urine pH, and patients converting to calcium phosphate had received more potassium citrate.
- Citrate in **calcium phosphate formers is controversial** (no RCTs): if used, **repeat the 24-hour urine** for citrate, pH and supersaturation. Patients whose **pH rises without a citrate rise** benefit less than those whose citrate rises without excessive alkalinity.

DIETARY INTERVENTION FOR HYPOCITRATURIA

- Citrus juices, chiefly lemonade and orange juice, supplement water to raise volume and citrate. Citrate is a weak organic acid that acts as a conjugate base when paired with sodium or potassium; absorbed citrate is almost entirely **metabolised to bicarbonate**.
- Citrate inhibits calcium oxalate stones by **forming soluble complexes with calcium** and impairing crystal nucleation and growth.
- Natural juices are richest in citrate and potassium: **grapefruit highest, then lemon and orange** (similar); Crystal Light is the richest commercial citrus drink.
- **Key distinction**: citrate in **orange and grapefruit** juice is complexed with **potassium** (an alkali load that raises pH), whereas in **lemon** juice it is **citric acid** (little alkali). **Malate** also raises urinary citrate. Therefore **alkaline citrate matters more than total citrate**.
- **Lemonade**: several studies showed rises in urinary citrate (one with smaller stone burden), cheaply and with good tolerance; at least three found no benefit, one showing potassium citrate succeeded where lemonade failed, another that lemonade helped only when combined with potassium citrate.
- **Orange juice**: less citrate than lemon but potassium-bound; associated with lower stone risk in three large cohorts and gives an alkali load and citrate rise comparable to potassium citrate — but it **raises urinary oxalate** and does not lower calcium, unlike potassium citrate.
- Other juices (lime, cranberry, black currant, grapefruit, tomato) give variable citrate rises.

- **Calories are the main drawback**, since many formers are overweight or diabetic; **lemonade is preferable** because its sweetener can be controlled. Low-calorie citrus sodas are citrate-rich; among sports drinks **Performance** raised citrate and pH but **Gatorade** did not, and both carry sugar.
- Over-the-counter citrate supplements are **not FDA evaluated or approved** (Table 99.3).
- **Summary:** no beverage is superior; lemonade has most evidence. **Volume matters more than type** — drink at least **3000 mL/day** for urine output above **2500 mL/day**.

TABLE 99.3 MEASURED PH, CITRATE, AND CITRATE AS ALKALI FOR COMMON OVER-THE-COUNTER FOOD PRODUCTS AND SUPPLEMENTS

Pattern	Products	Features
Almost all citrate delivered as alkali	Litholyte Powder, Litholyte Coffee, NOW Potassium Citrate, Horbaach Potassium Citrate, TheraLith XR	Neutral to alkaline pH (about 6.4–10.8); modest citrate per serving (about 2.5–12.6 mEq)
High total citrate, much of it acid	Moonstone Powder, LithoBalance, KSP tabs	Acidic pH (about 3.4–4.8); 27–64 mEq total citrate but only about 12–16 mEq as alkali
Low content	Kidney COP capsule	pH about 5.4; 5.5 mEq total, 3.5 mEq alkali

Values are averages of two lots. Modified from Dai JC, Maalouf NM, Hill K, Antonelli JA, Pearle MS, Johnson BA: Alkali citrate content of common over-the-counter and medical food supplements, J Endourol 37(1):112–118, 2023.

- **What the table is for:** total citrate on a label does not predict urinary effect; the **alkali citrate** column does.
- **Trap:** the product with the **highest total citrate** (Moonstone) is strongly acidic and delivers only a fraction as alkali — the supplement equivalent of lemon juice versus orange juice.

CITRUS AND CITRUS-BASED BEVERAGES COMPARED

Beverage	Citrate form	Effect on urine	Drawback
Lemonade	Citric acid	Citrate rise inconsistent	Most evidence; calories controllable — preferred
Orange juice	Potassium citrate	pH and citrate rise comparable to potassium citrate	Raises oxalate; no fall in calcium; calories
Grapefruit juice	Potassium-bound	Highest total citrate	Calories
Low-calorie citrus sodas, Crystal Light	—	High citrate content	—
Performance vs Gatorade	—	Performance raises citrate and pH; Gatorade no effect	Fructose and calories

Concept — *urinary citrate is raised by an alkali load, not by citrate as such — so potassium citrate is the reference agent, alkali-rich juices outperform acidic ones, and over-alkalinisation trades calcium oxalate risk for calcium phosphate risk.*

URIC ACID STONES

- **Urine pH is the most important determinant** of uric acid solubility; with **no known crystallisation inhibitors**, undissociated uric acid precipitates once urine is supersaturated.
- Sigmoid solubility curve: **pKa about 5.5** (50% soluble); **at pH 6.5 over 90% is ionised** and soluble. **Urine pH below 5.5 is the commonest cause** of uric acid stones.

- Low urine pH has two mechanisms: **increased net acid excretion** and **reduced renal ammonium excretion**.

Dietary Intervention for Uric Acid Stone Prevention

- Fluid for **urine volume of at least 2.5 L/day**.
- With hyperuricosuria, **limit nondairy animal protein** — dietary purines account for about **30% of urinary uric acid**; eat fruit and vegetables for their alkali load.
- Exercise, weight loss and glucose control help in obesity, metabolic syndrome and diabetes.

Pharmacologic Intervention for Uric Acid Stone Prevention

- Goal: urine pH **above 5.5, preferably 6.0–6.5**.
- **Sodium alkali** raises pH but can cause **calcium-containing stones**; **potassium citrate** alkalinises without that complication and is preferred, at **30–60 mEq/day in two or three doses** to keep pH about 6.5. **Avoid pH above 7.0** (calcium phosphate risk).
- **Allopurinol is not routine first-line**, because low urine pH is by far the dominant risk. Exceptions: **hyperuricaemia with hyperuricosuria** (e.g., primary gout); AUA also allows it as an adjunct when **alkalinisation fails** or stones recur despite it.
- Allopurinol: **300 mg/day**; renally cleared, so **reduce the dose in renal impairment**. Common effects: rash, muscle pain, nausea, vomiting, diarrhoea.
- In **2–8%**, reactions range from cutaneous hypersensitivity to **allopurinol hypersensitivity syndrome (AHS), fatal in up to 27%**; allopurinol is a common cause of **Stevens-Johnson syndrome** and complicated renal failure. Risk rises with higher dose, renal insufficiency, diuretics and statins, and with **concurrent thiazide in reduced renal function**.

Concept — *uric acid stones are a pH disease more than a uric acid disease: alkalinise to 6.0–6.5 with potassium citrate, and reserve allopurinol for true hyperuricosuria or failed alkalinisation.*

CYSTINE STONE PREVENTION

Dietary Interventions for Cystine Stone Prevention

- Two goals: **lower cystine concentration below 250 mg/L** (hydration, diet) and **raise cystine solubility** (alkalinisation, binding drugs).
- High fluid is **absolutely essential**: urine output of **at least 3 L/day, spread over 24 hours including the night**, to prevent physiological nocturnal concentration. Intake 3–5 L/day for adults and 3 L/day for children, increased with exercise, fever and heat.
- **Sodium restriction** reduces cystine excretion; restriction to 1 mmol/kg/day also increases proximal tubular cystine reabsorption. Target **sodium of 2300 mg/day or less**. Alkalinising with **sodium** rather than potassium citrate reduces the efficacy of tiopronin.
- **Protein restriction**: animal protein supplies cystine and **methionine** (metabolised to cystine), and a low-protein diet lowers cystine excretion — but **not in children and adolescents**, as methionine is essential for growth and scarce in plant foods; no effect on recurrence has been shown.
- AUA: counsel **limited sodium and protein**; encourage fruit and vegetables rich in malate and citrate.

Pharmacologic Intervention Prevention of Cystine Stones

- **First line** (with hydration and diet): alkalinise to **pH 7.0–7.5**, preferably with **potassium citrate 30–60 mEq/day** titrated to pH, since sodium increases cystine excretion.

- Cystine's pKa of **8.3** is impractical to reach, and **pH above 7.5 risks calcium phosphate** stones.
- Potassium citrate causes nausea, pain, vomiting and diarrhoea (less with meals); use with caution in **impaired GI motility** (gastric ulceration) and in **renal impairment or with angiotensin-converting enzyme (ACE) inhibitors** (hyperkalaemia).
- **Second line — thiols**: a sulfhydryl group forms a **soluble mixed disulfide** (cystine–drug in place of cystine–cystine).

THIOL (CYSTINE-BINDING) DRUGS COMPARED

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol (1963); largely replaced	FDA approved 1988; thiol of choice	ACE inhibitor; enthusiasm waned
Evidence	Added to conservative therapy, reduced recurrence, passage and growth; thiols cut yearly stone events by about two-thirds	Possibly more effective; far fewer patients stop it than D-penicillamine	Disulfide about 200 times more soluble in vitro; case reports, contradictory results, no long-term trials; urinary excretion may be too low
Dose	Start 250 mg/day, titrate	Start 300 mg twice daily, up to 1200 mg/day; titrate to cystine below 250 mg/L	—
Toxicity	Rash, fever, arthralgia, pancytopenia, proteinuria, nephrotic syndrome ; some long-term users tolerate it	Asthenia, GI upset, rash, joint aches, mental status changes; serious reactions less common	—

Concept — *cystinuria is managed as a solubility equation: dilute (at least 3 L of urine around the clock, less sodium and protein), then alkalinise to 7.0–7.5 with potassium, then chemically split cystine with a thiol (tiopronin).*

URINE PH TARGETS AND CHOICE OF ALKALI BY STONE TYPE

Stone	Target urine pH	Preferred alkali	Why not sodium alkali
Uric acid	6.0–6.5; avoid above 7.0	Potassium citrate 30–60 mEq/day	Calcium-containing stones
Cystine	7.0–7.5; avoid above 7.5	Potassium citrate, titrated	Raises cystine excretion; blunts tiopronin
Calcium oxalate with hypocitraturia	Alkaline, below 7	Potassium citrate	Sodium citrate does not lower urinary calcium
Calcium phosphate	Avoid high pH; recheck 24-hour urine	Controversial	—

PREVENTION OF INFECTION STONE RECURRENCE

- Infection (**struvite**) stones are treated by **aggressive surgery aiming at complete clearance**; the AUA strongly recommends endoscopic surgery — **percutaneous nephrolithotomy (PCNL) first line for complex staghorn calculi** — because **every infected fragment must be removed** to prevent recurrence (Chapter 98).
- Medical management aims at **preventing recurrence, not dissolution**: long-term control of **urea-splitting** infection, better bladder health, adequate drainage and surveillance for reinfection.
- Control is hard with **residual fragments**, which harbour organisms and endotoxin; recolonisation is likely even after cultures turn negative.

- Antibiotics follow culture, but urine and stone cultures correlate poorly — so all removals of presumed infection stones need **broad-spectrum cover** suited to local resistance.

Acetohydroxamic acid

- AHA is an oral **urease inhibitor** that reduces struvite saturation, concentrates in urine and penetrates bacterial walls; at **250 mg three times daily** it prevents new stones and slows growth, as shown in placebo-controlled RCTs (including in spinal cord injury), and occasionally dissolves stones.
- AUA: offer AHA for residual or recurrent struvite stones **only after surgical options are exhausted** (Grade B).
- Side effects in up to **30%**; **haemolytic anaemia in 3–15%** requires stopping; the most severe is **deep venous thrombosis**, from low-grade intravascular coagulation. Trial withdrawal rates were high.
- **Contraindications:** creatinine above **2.5 mg/dL**, pregnancy, and childbearing age without contraception; renal insufficiency increases toxicity. It is therefore reserved mainly for patients **unfit for surgery**.

Metabolic evaluation in struvite formers

- Controversial: **mixed struvite–calcium oxalate** formers have more metabolic abnormalities than **pure struvite** formers, yet most of both groups have abnormalities — commonly **hypercalciuria and hypocitraturia**. Treating these along with the infection (AHA and/or suppressive antibiotics) prevented growth of residual stones in most, so a 24-hour urine is worthwhile.

Concept — *struvite is a surgical disease first: every fragment is an infected nidus, drugs only hold the line, and the toxicity of AHA confines it to the patient who cannot have surgery.*

MONITORING PREVENTIVE STRATEGIES

- AUA: one **24-hour urine within 6 months** of starting drug therapy, then at least **annual** evaluation for adherence and response, with **blood tests** for adverse effects.
- If stones recur on therapy, **repeat stone analysis** to detect an evolving metabolic condition.
- Periodic **imaging** for silent growth or new stones, balanced against radiation exposure (expert opinion).

KEY POINTS

- Dietary prevention measures, including high fluid intake, form the foundation for preventive measures, even among individuals with high-risk stone disease.
- Effective pharmacotherapeutic options exist for hypocitraturia and hypercalciuria.
- RNA interference represents a recent breakthrough for individuals with PH1.

Concept — *prevention is a loop: test, treat the specific abnormality, re-test within 6 months and then yearly, and re-analyse any new stone.*